ETS Decomposition

Renato R. Maaliw III, *DIT*
College of Engineering
Southern Luzon State University
Lucban, Quezon, Philippines

ETS

- stands for Error, Trend, and Seasonality
- a method used to break down a time series into three components

Trend (T)

- Overall direction in which the time series is moving (upward, downward, stable/flat)

Seasonality (S)

- Repeating patterns in the series (e.g. daily, monthly, yearly)

Residual/Error (E)

- The noise or irregular fluctuations that cannot be explained by trend or seasonality

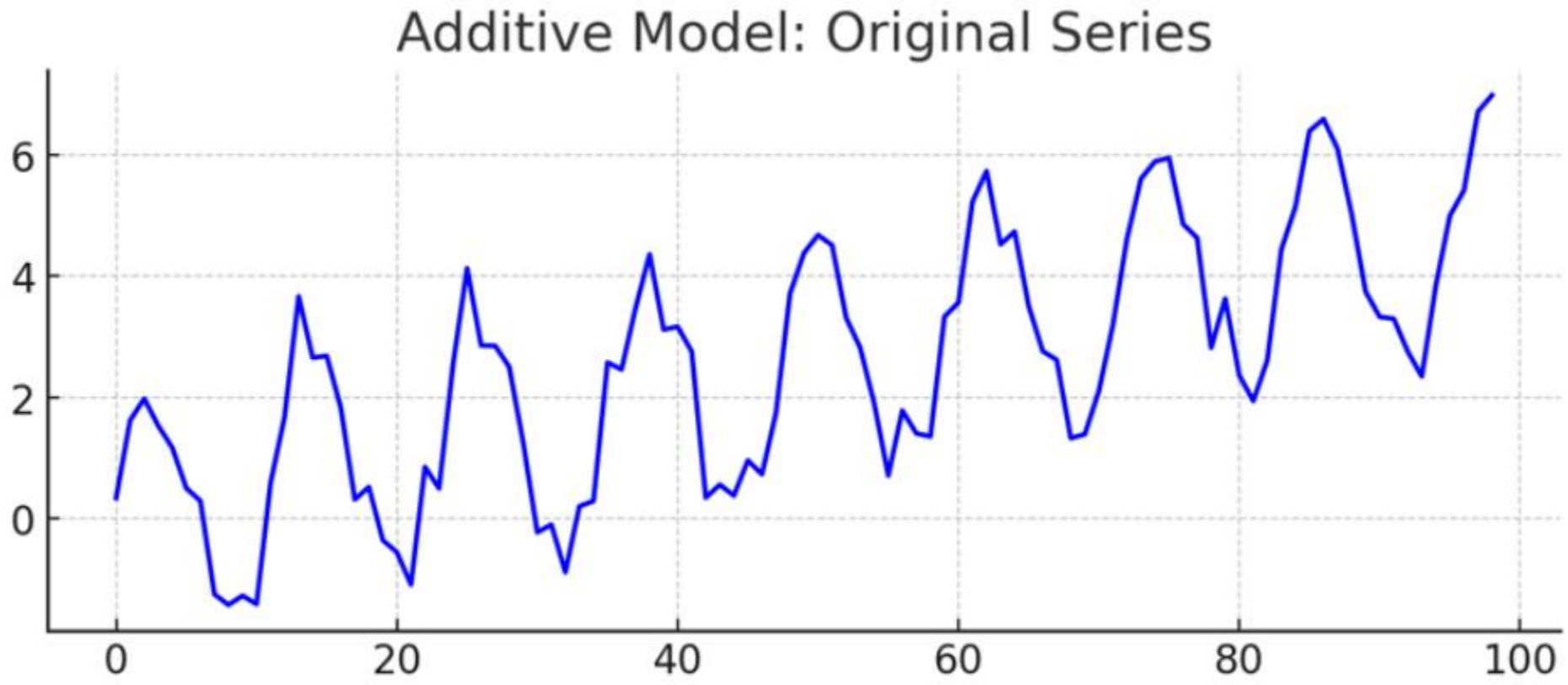
ETS decomposition allows us to understand the underlying patterns in a time series and helps in forecasting

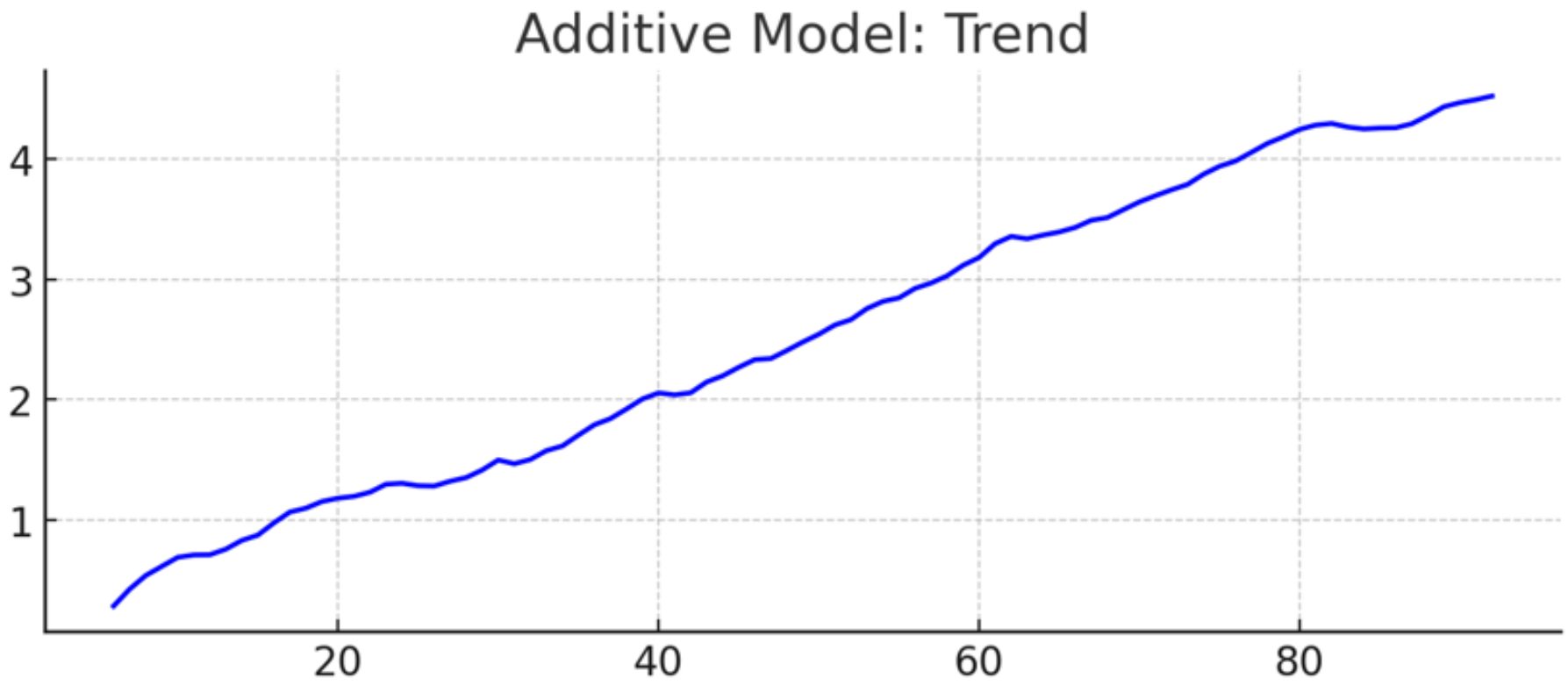
Types of ETS Models

1. Additive Model

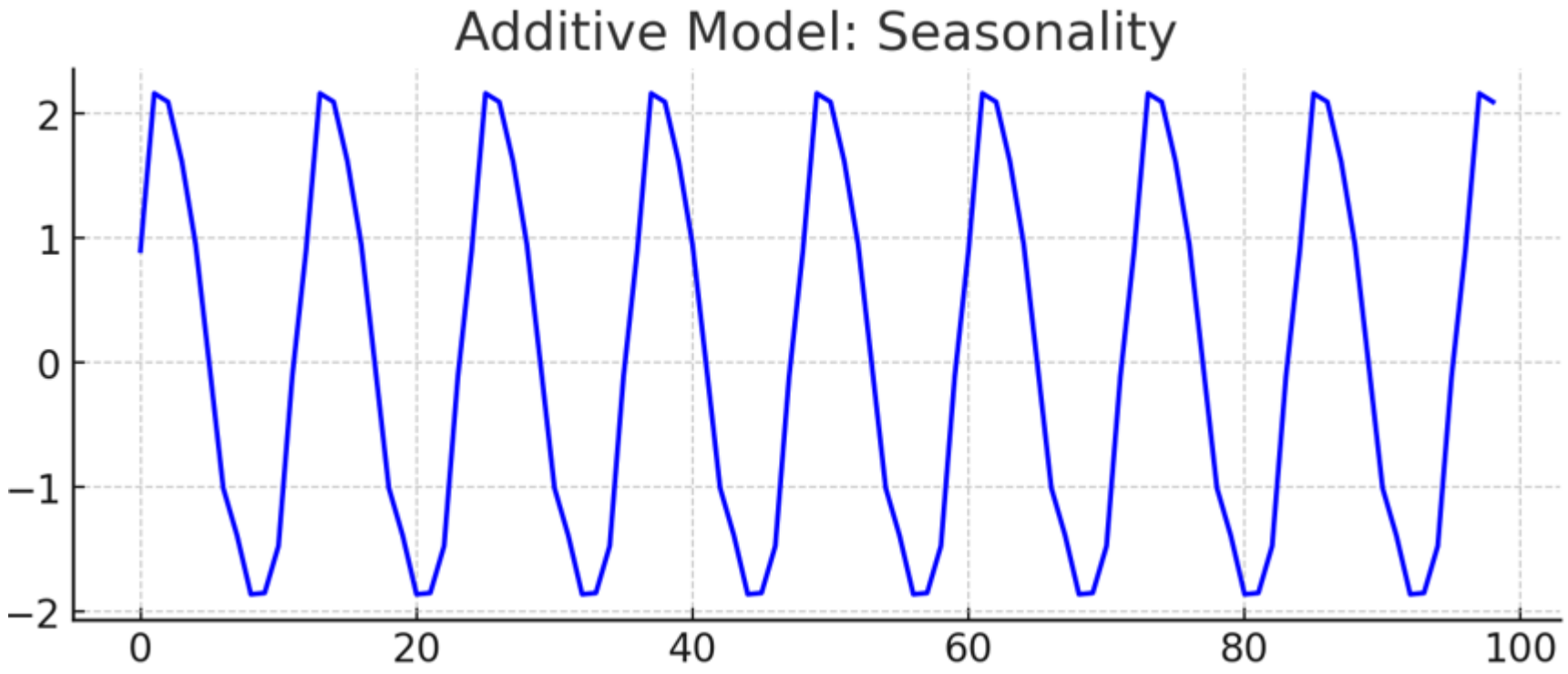
   - when seasonal variations remain constant over time

Cognate/Professional Electives

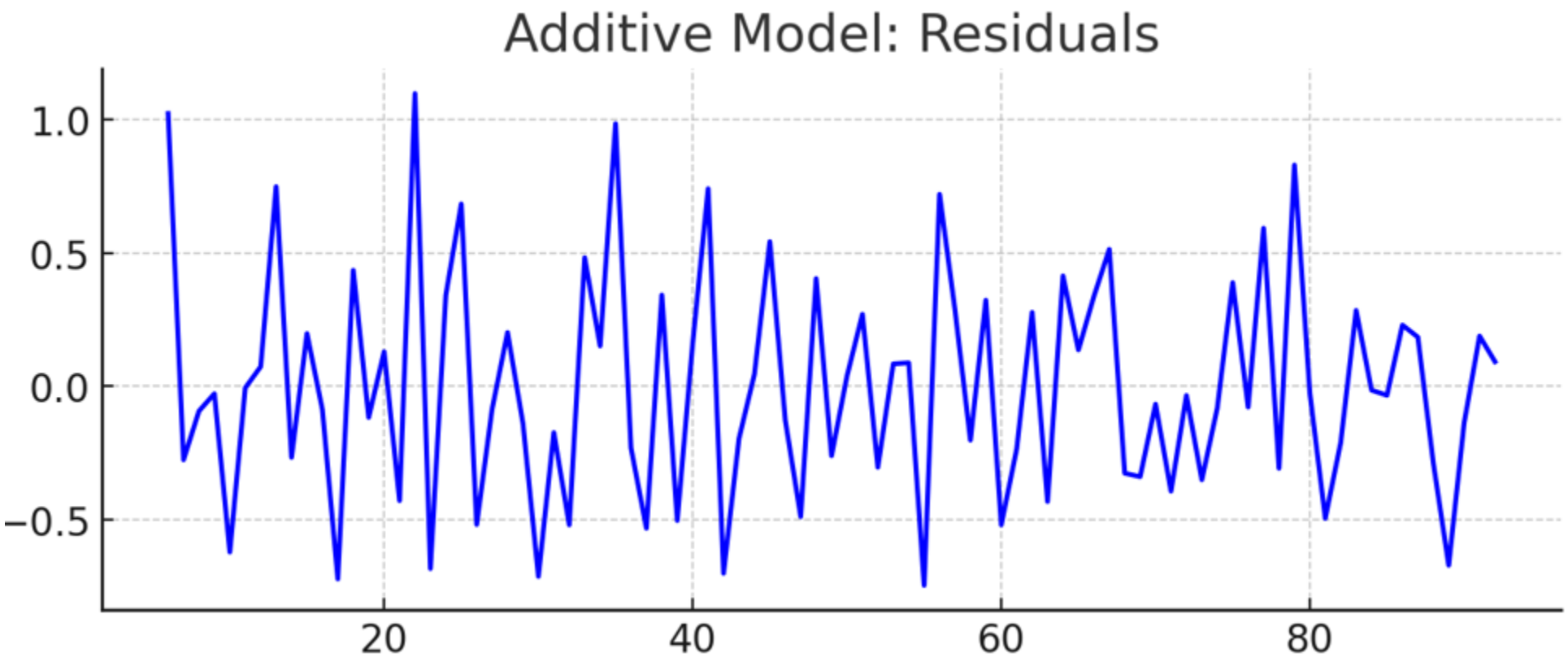




Additive Model's trend increase linearly over time



Additive Model's seasonal pattern remains constant in magnitude (fixed oscillations)



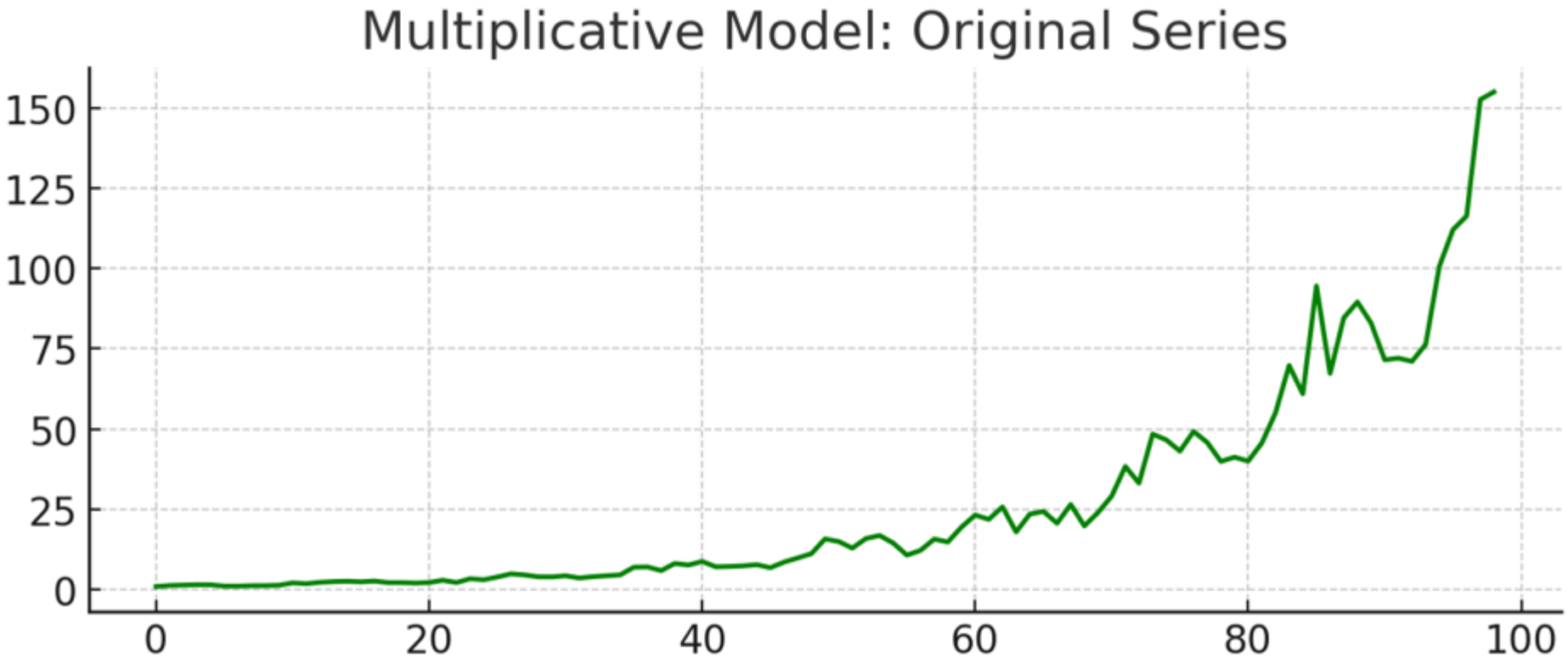
Additive Model's residuals have constant noise over time

Types of ETS Models

2. Multiplicative Model

   - when seasonal variations increase or decrease in size over time

Cognate/Professional Electives



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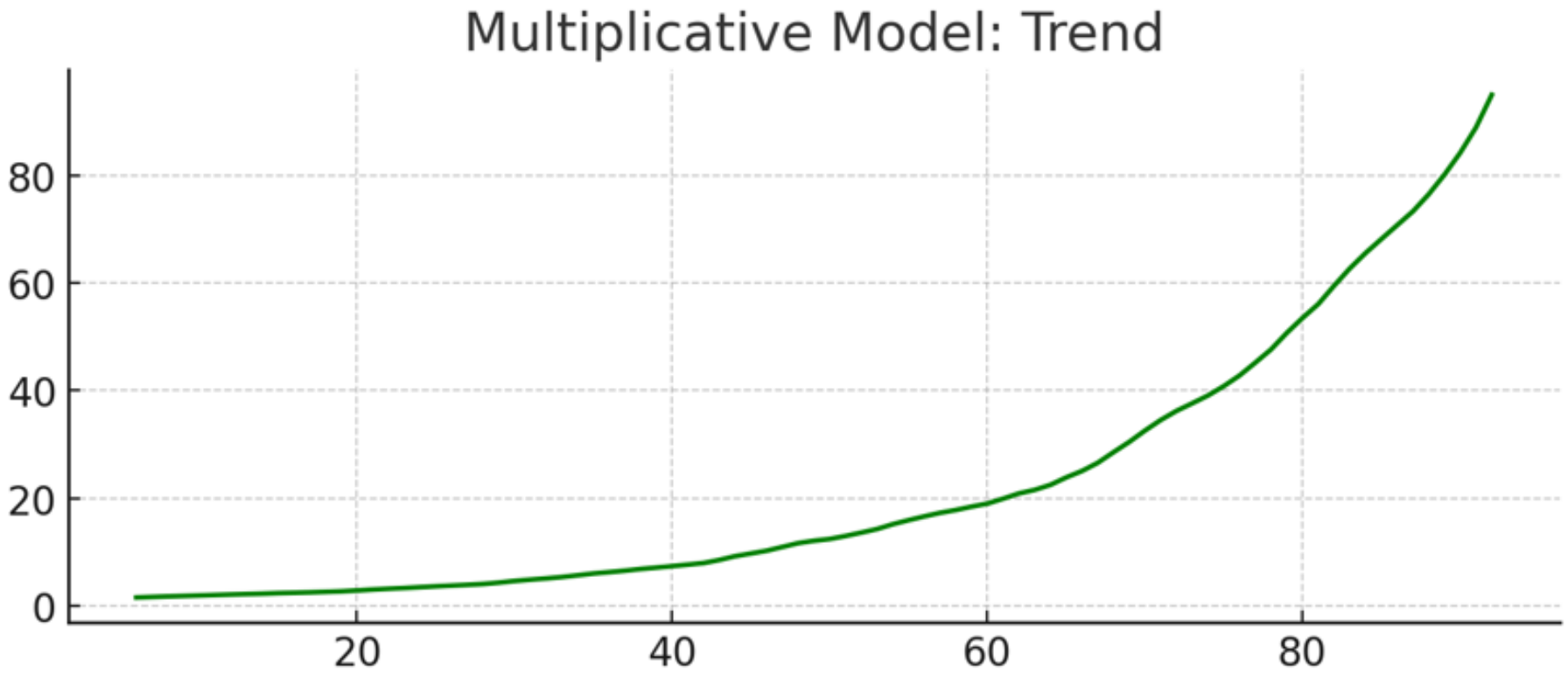
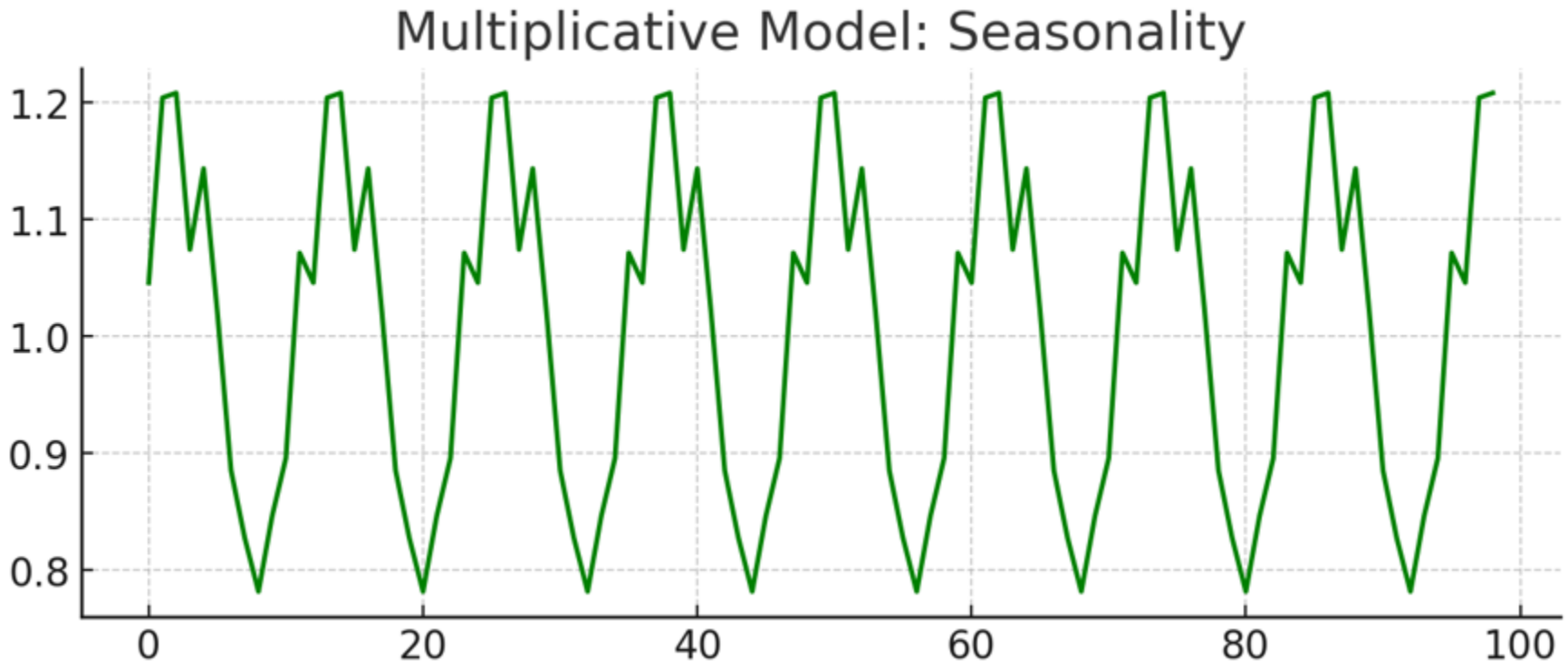
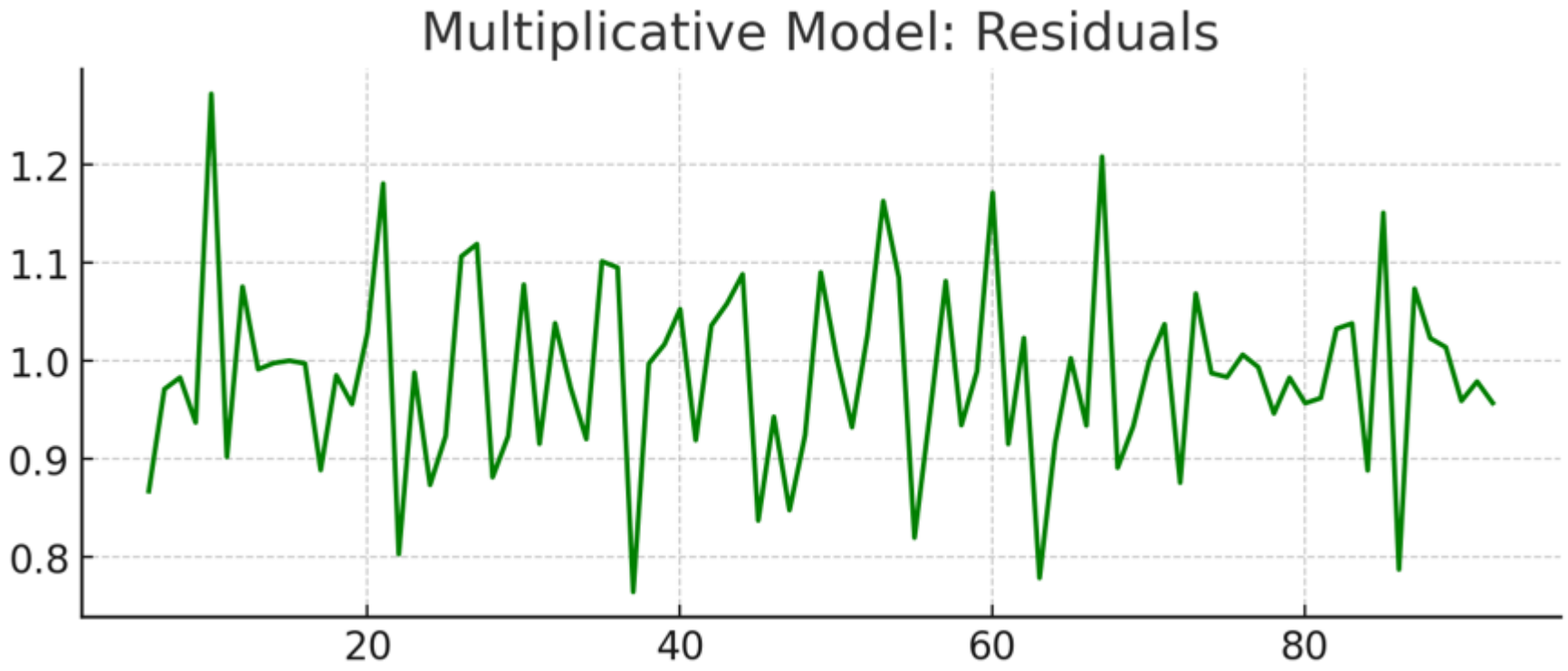


Multiplicative Model’s trend grows exponentially



Multiplicative Model's seasonal pattern increases in magnitude over time (amplitude grows larger)



Multiplicative Model's residuals noise grows larger as the series increases

[Code Demo]

01. What does ETS stand for in time series decomposition

A. Exponential Time Series
B. Error Trend Seasonality
C. Expert Time Series
D. Error Trial Seasonality

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B. Inflation
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03. How does seasonality differ from cyclical patterns?

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B. Seasonality involves consistent period, cyclic behavior does not
C. Seasonality is driven by static factors, cycles are driven by changes
D. Both are identical; they can be used interchangeably

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05. When would you choose a multiplicative model over an additive one?

A. Seasonal fluctuations remain constant regardless of trend
B. Seasonal fluctuations grow proportionally as trend increases
C. The series contains cyclical patterns
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Thank you very much for listening.